Problem G. Same Sum

Time limit: 2 seconds
Memory limit: 1024 megabytes

Bobo is working with an integer sequence $a_1, a_2, \ldots, a_n$ of length n. He must process q queries in order. Each query is of one of the following two types:

- 1 L R v ($1 \le L \le R \le n$, $0 \le v \le 2 \cdot 10^5$): for all $i \in [L, R]$, update $a_i \leftarrow a_i + v$;
- 2 L R ($1 \le L < R \le n$, $R - L + 1$ is even): determine if elements $a_L, a_{L+1}, \ldots, a_R$ can be divided into $(R - L + 1)/2$ pairs of integers with the same sum.

Your task is to help Bobo process these queries efficiently.

Input

The first line of input contains two integers n, q ($1 \le n, q \le 2 \cdot 10^5$).

The second line of input contains n integers $a_1, a_2, \ldots, a_n$ ($0 \le a_i \le 2 \cdot 10^5$).

Then q lines follow. Each of the following lines contains a query, described in the statement.

Output

For each query of the second type, output "`YES`" (without quotes) in one line if elements $a_L, a_{L+1}, \ldots, a_R$ can be divided into $(R - L + 1)/2$ pairs of integers with the same sum; otherwise, output "`NO`" (without quotes) in one line.

You can output "`YES`" and "`NO`" in any case (for example, strings "`yES`", "`yes`", and "`Yes`" will all be recognized as a positive response).

Example

standard input	standard output
8 4 1 2 3 4 5 6 7 8 2 1 8 1 1 4 4 2 1 6 2 1 8	YES NO YES